

1. Project Understanding

User Prompt:

Explain the Django assignment and requirements.

AI Response:

The project is a Django REST Framework based Box Selection System that recommends the most suitable shipping box for an ecommerce order based on weight, dimensions, and volume. It uses SQLite database and includes models for Products, Boxes, Orders, and Order Items.

2. Project Setup

User Prompt:

Help me set up Django project and run server.

AI Response:

```
python -m venv .env
.venv\Scripts\activate
pip install -r requirements.txt
python manage.py migrate
python manage.py runserver
```

3. Box Selection Logic

User Prompt:

How does box recommendation work?

AI Response:

The system checks:

- Total order weight $\leq$ box max weight
- Product dimensions fit inside box
- Total product volume $\leq$ box volume

From all valid boxes, it selects the lowest cost box. If cost is same, smaller box is selected.

This is a rule-based system, not a 3D packing algorithm.

4. Django Models

User Prompt:

Explain models in project.

AI Response:

- Product $\rightarrow$ name, length, width, height, weight
- Box $\rightarrow$ dimensions, max weight, cost, is_active

- Order → customer_name
 - OrderItem → product, quantity
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5. API Endpoints

User Prompt:

Create REST APIs.

AI Response:

- /api/products/
 - /api/boxes/
 - /api/orders/
 - /api/orders/<id>/recommend-box/
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6. Recommendation API

User Prompt:

Explain recommend-box API.

AI Response:

It filters active boxes and checks constraints (weight, dimensions, volume). It returns the lowest cost suitable box. If no box matches, returns error message.

7. Git & GitHub Setup

User Prompt:

How to push project to GitHub?

AI Response:

```
git init
git add .
git commit -m "Initial commit"
git remote add origin <repo-url>
git push -u origin main
```

8. Testing

User Prompt:

How to run tests?

AI Response:

```
python manage.py test
```

Expected Output:

Ran 9 tests in 0.164s
OK

9. Git Ignore Explanation**User Prompt:**

Why .venv not pushed?

AI Response:

.venv is a local environment folder and is ignored using .gitignore because it is large and not needed in repository.

10. AI Tools Used**User Prompt:**

Which AI tools were used?

AI Response:

Codex and ChatGPT were used to assist in development, including project structure, API design, and logic implementation. All outputs were reviewed and manually adjusted.

11. What I Learned**AI Response:**

- Django project structure and MVC pattern
 - Django REST Framework APIs
 - Database modeling using SQLite
 - Business logic for shipping optimization
 - Git and GitHub workflow
 - Using AI tools for development assistance
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12. Final Verification**User Prompt:**

Check project readiness.

AI Response:

Verified:

- Django structure correct
- APIs working

- Tests passing
- GitHub setup correct
- Documentation complete